Assembly Guide SO-ARM101

This guide provides step-by-step instructions for assembling the SO-ARM101. Follow these steps carefully to ensure proper operation.

Required Tools:

- Phillips head screwdriver

- Small pliers (for handling cables)

Step 1: Servos preassembly

Components needed:

- 5x STS3215

- 5x Servo Cables (from servo kit)

- 5x Motor Horns (from servo kit)

- 4x Passive Horns (from servo kit)

- 5x M3x6 Screws (from servo kit)

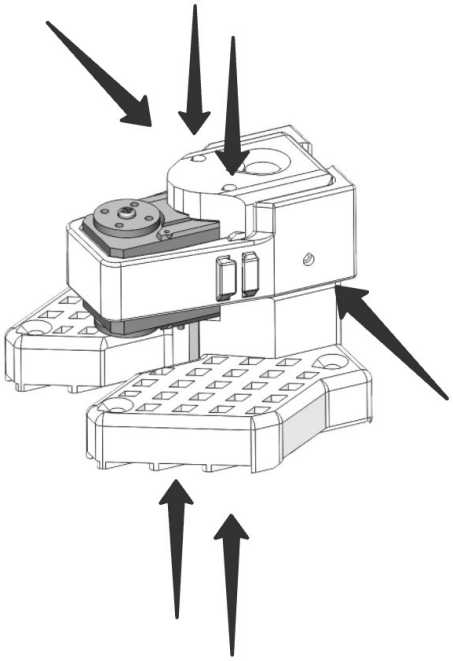
- 4x Long self-tapping screws (from servo kit)

Repeat next steps for each servo.

1. Install servo horn, secure with M3x6 screw.

2. Install passive horn, secure with self-tapping screw. Make this step only for 4x servos, one leave with only a servo horn. This servo will be needed for the very last wrist joint.

3. Connect one 3-pin cable.



Step 2: Base assembly

Components needed:

- 1x Base_SO101

- 1x Base_motor_holder_SO101

- 1x STS3215

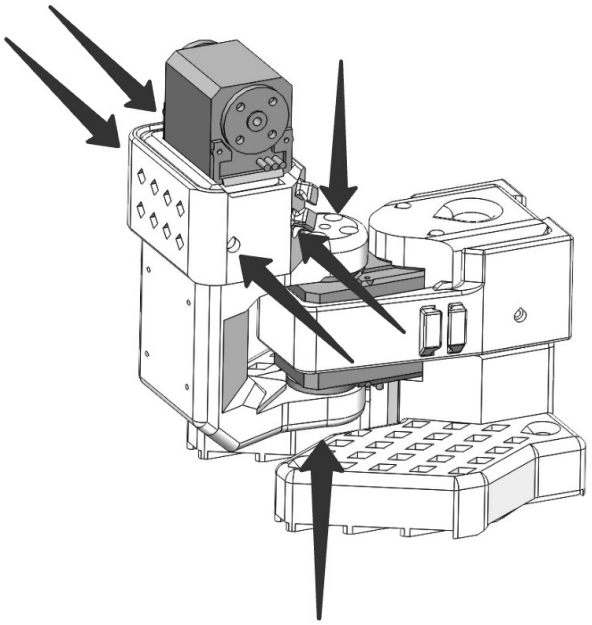
- 6x Self-tapping screws (from servo kit)

1. Place the first servo into the Base_SO101 (cable looking down).

2. Fasten the servo with 4 self-tapping screws (smallest screws). Two from the top and two from the bottom.

3. Slide over the first servo Base_motor_holder_SO101 and fasten it using two self-tapping screws (one on each side).

Step 3: Waist Joint Assembly



Components needed:

- 1x Rotation_Pitch_SO101
- 1x Motor_holder_SO101_Base
- 1x STS3215
- 8x M3x6 Screws (from servo kit)
- 4x Self-tapping screws (from servo kit)

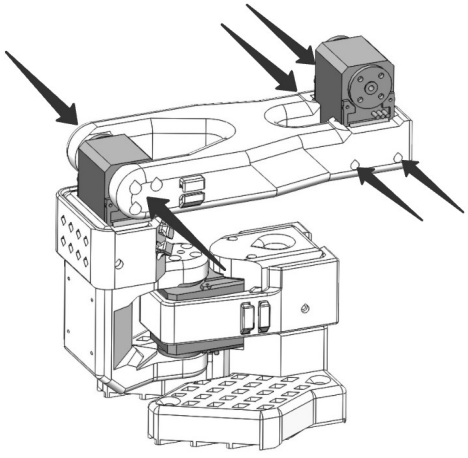
1. Attach the Rotation_Pitch_SO101 part.

2. Tighten it with 4 M3x6mm screws on top and 4 M3x6mm screws on the bottom.

3. Add the Motor_holder_SO101_Base.

4. Slide the second servo in from the top (pay attention that cable is looking at us).

5. Fasten the second servo with 4 self-tapping screws.



Step 4: Shoulder Joint Assembly

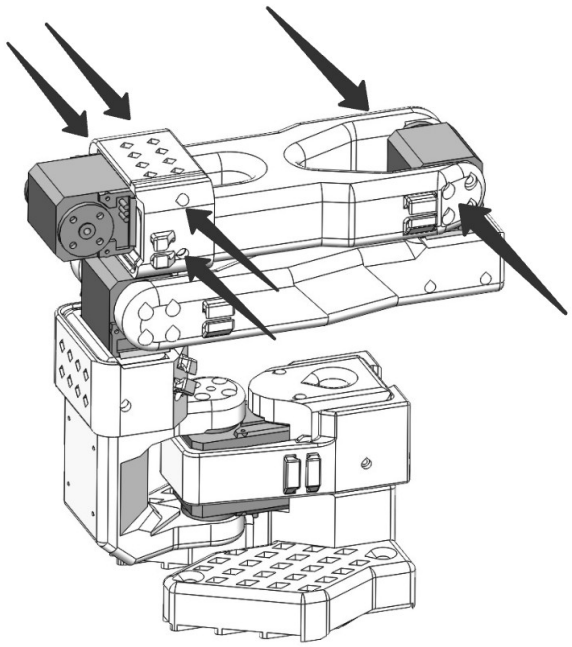
Components needed:

- 1x Upper_arm_SO101
- 1x STS3215
- 8x M3x6 Screws (from servo kit)
- 4x Self-tapping screws (from servo kit)

1. Attach the Upper_arm_SO101 with 4 M3x6mm screws on each side.

2. Insert servo №3 and fasten using 4 self-tapping screws (pay attention that cable is looking at us).

Step 5: Forearm Joint Assembly



Components needed:

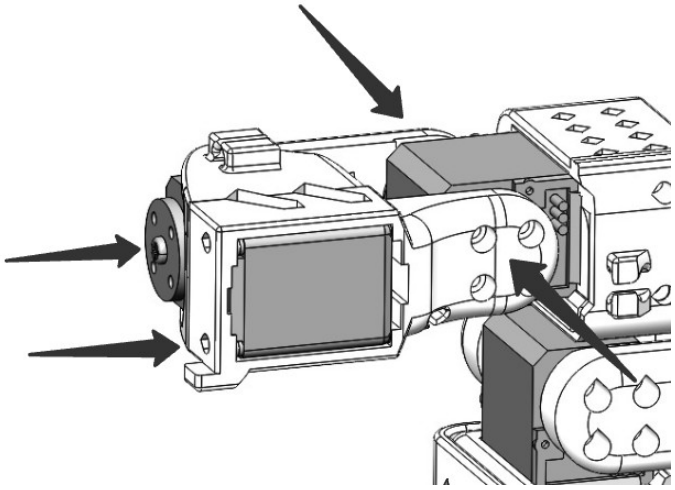
- 1x Under_arm_SO101
- 1x Motor_holder_SO101_Wrist
- 1x STS3215
- 8x M3x6 Screws (from servo kit)
- 4x Self-tapping screws (from servo kit)

1. Attach the Under_arm_SO101 with 4 M3x6mm screws on each side.

2. Add the Motor_holder_SO101_Wrist.

3. Insert servo №4 and fasten using 4 self-tapping screws (pay attention that cable is looking at us).

Step 6: Wrist Joint Assembly



Components needed:

- 1x Wrist_Roll_Pitch_SO101
- 1x STS3215
- 8x M3x6 Screws (from servo kit)
- 2x Self-tapping screws (from servo kit)

1. Insert servo №5 into Wrist_Roll_Pitch_SO101 and fix it with 2x Self-tapping screws.
2. Attach this assembly and fix it with 4 M3x6mm screws on each side.

Step 7: Existing Arm Controller

Use the controller and power supply already installed on your SO-ARM101.
Follow the controller manufacturer instructions for mounting and power.
The controller and power supply belong to the arm setup and are outside
the parallel gripper bill of materials.

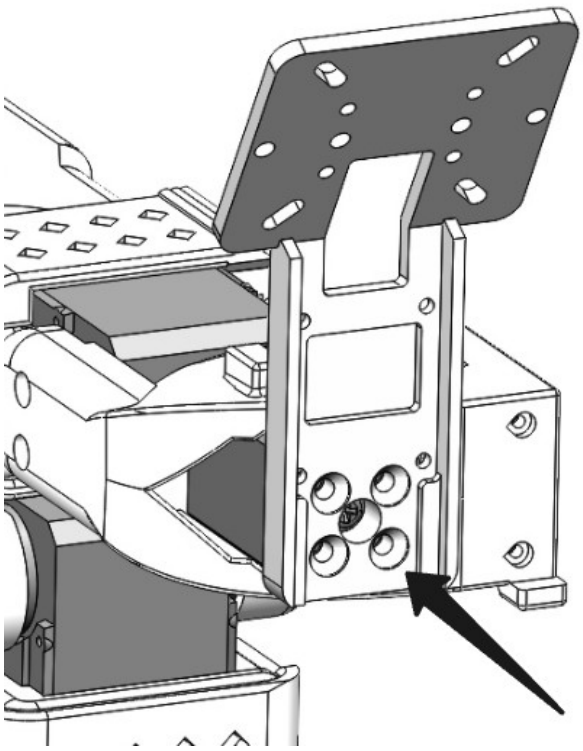
Step 8: Servo Bus Connections

1. Switch off arm power before connecting servo cables.
2. Connect the arm servos to each other in a chain.
3. Connect the first servo to the existing arm controller servo bus.
4. Connect the gripper cable to servo 5 (see Step 9).
5. Ensure every servo has a unique ID before using the shared bus.
6. Connect the controller to the PC by USB and use its serial port.
   Close other control applications before opening that port.

Connection Overview

PC (USB) -> Existing arm controller -> Arm servo bus -> Gripper
Power: use the existing arm supply configured for the installed servos.

Step 9: Gripper Holder Attachment



Components needed:

- 1x Camera holder (RB9.01.060.071)

or Servo plate (RB9.01.060.080)

- 1x Parallel Gripper SO-ARM101

- 4x M3x6 Screws (from servo kit)

- 4x Self-tapping screws (from servo kit)

1. Depending on what type of Gripper you want to connect attach to the Wrist Joint horn Camera holder or Servo plate and fix with 4 M3x6 screws.

2. Attach the Parallel Gripper to the Camera holder or Servo plate and fix with 4 self-tapping screws.

3. Connect cable from the Parallel Gripper to the servo №5 .

Your SO-ARM101 is now assembled.